

Sri Lanka Institute of Information Technology



Lab sheet No 05

Exercise 05

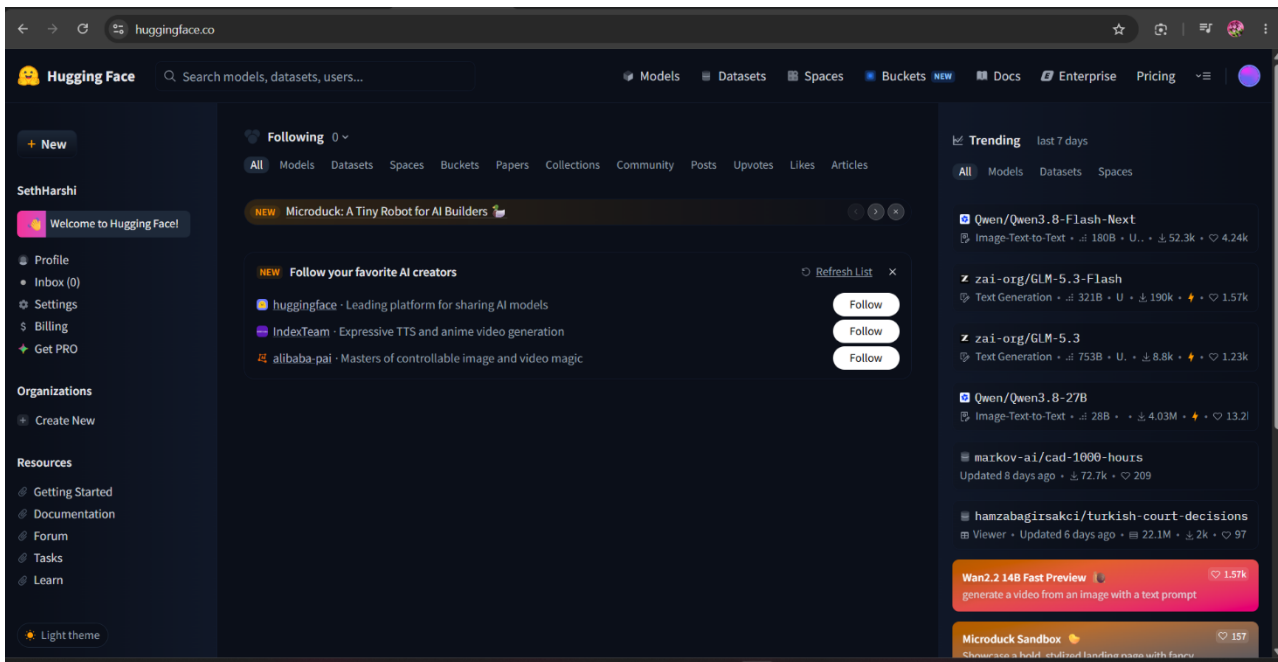
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Udumullage S.H.U.

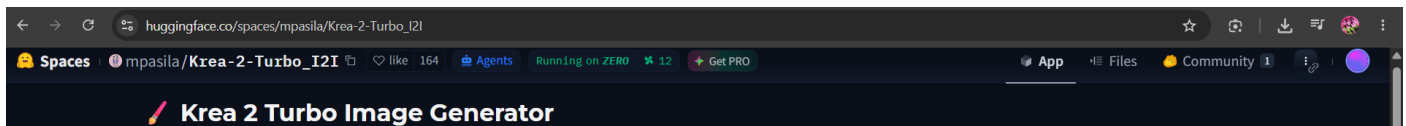
IT2011 - Artificial Intelligence and Machine Learning

B.Sc. (Hons) in Information Technology

1. Create an account in <https://huggingface.co>

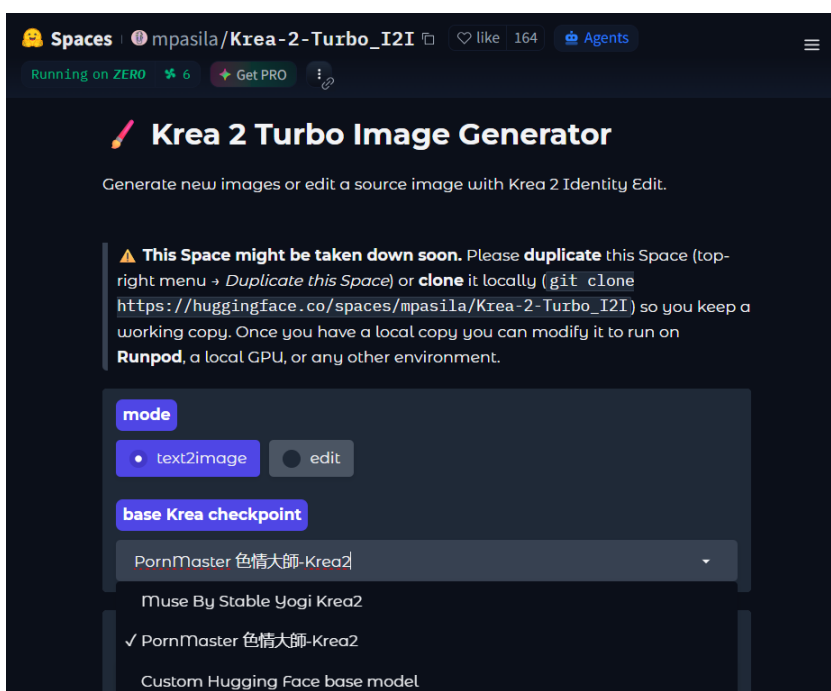


2. Select any AI project of your liking under Spaces



3. Try out the AI project and generate the output

https://huggingface.co/spaces/mpasila/Krea-2-Turbo_I2I



prompt

A breathtakingly beautiful natural landscape at golden hour, a peaceful crystal-clear lake surrounded by lush green mountains and wildflowers, soft warm sunlight shining through the trees, realistic reflections in the water, gentle mist floating above the lake, a few birds in the distant sky, natural colors, photorealistic, highly detailed, cinematic lighting, authentic untouched nature, realistic textures, serene and peaceful atmosphere, professional nature photography, 4K quality.

width

1024



height

1024



512



2048

512



2048

LoRAs (18 available)



sampling



seed



randomize seed

2

settings

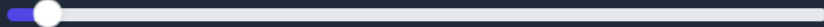


CPU budget (0 = automatic)

10



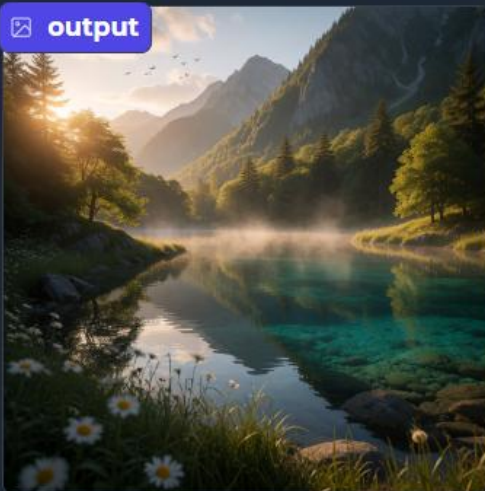
0

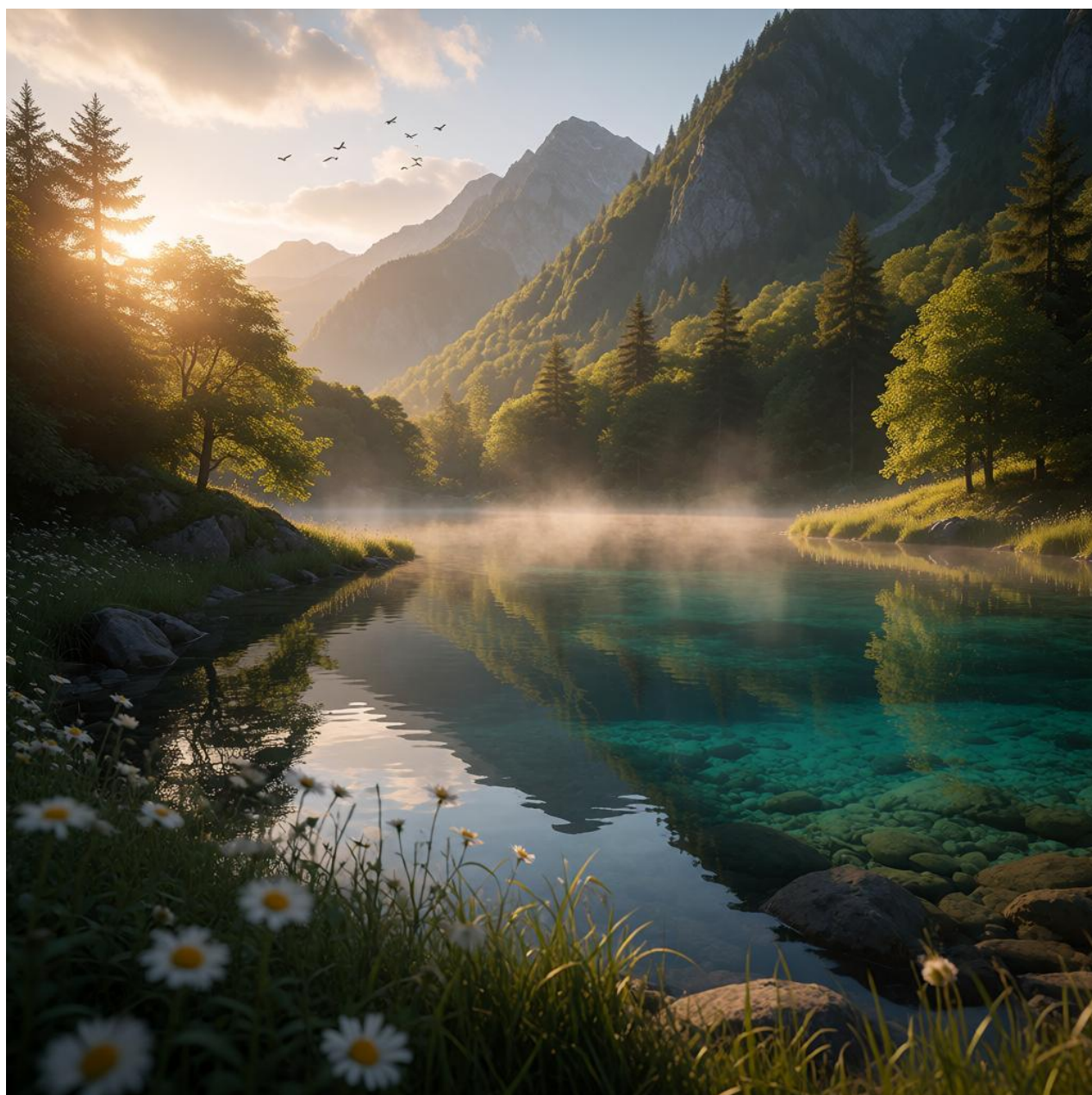


300

generate

output





status

done — 1 image(s), seed 3531208244

used seed

3531208244

5. Under Models select a Model of your choice to find more details about it



Omni-Video-Factory

Please give us a ❤️ if you find it helpful. No usage limits beyond your ZeroGPU daily quota.



Image to Video



Text to Video



Video Extend



Omni-Video-Factory

Industrial-grade creator workflow on ZeroGPU

What this project is

Omni-Video-Factory is an OmniAILab experimental integration that combines **OmniVideo**, **ComfyUI workflows**, and **Hugging Face ZeroGPU** to deliver industrial-grade video creator tooling in a lightweight public demo.

About OmniVideo

OmniVideo was first released in **December 2025** as a multi-modal generation and editing model for video and image. It has strong compatibility with ComfyUI and is designed for high-fidelity, identity-consistent video generation.



Open Source & Transparency

Comfy runtime code is based on minimal modifications of [selftcamera/ComfyUIVideo](#).

Except for OmniVideo model weights, the end-to-end pipeline code is visible.

Our [workflows](#) are also highly compatible with LTX-Video, WanX 2.2, and WanX 2.1. You can use AI coding tools to adapt this stack to corresponding open-source models.



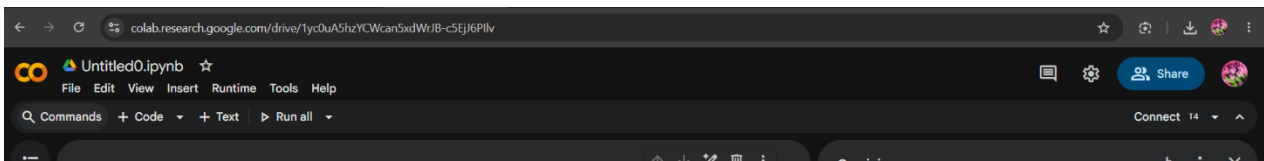
Access & Usage Policy

This demo does not enforce a custom generation count cap. Your available runs are determined by your Hugging Face account quota and ZeroGPU availability.

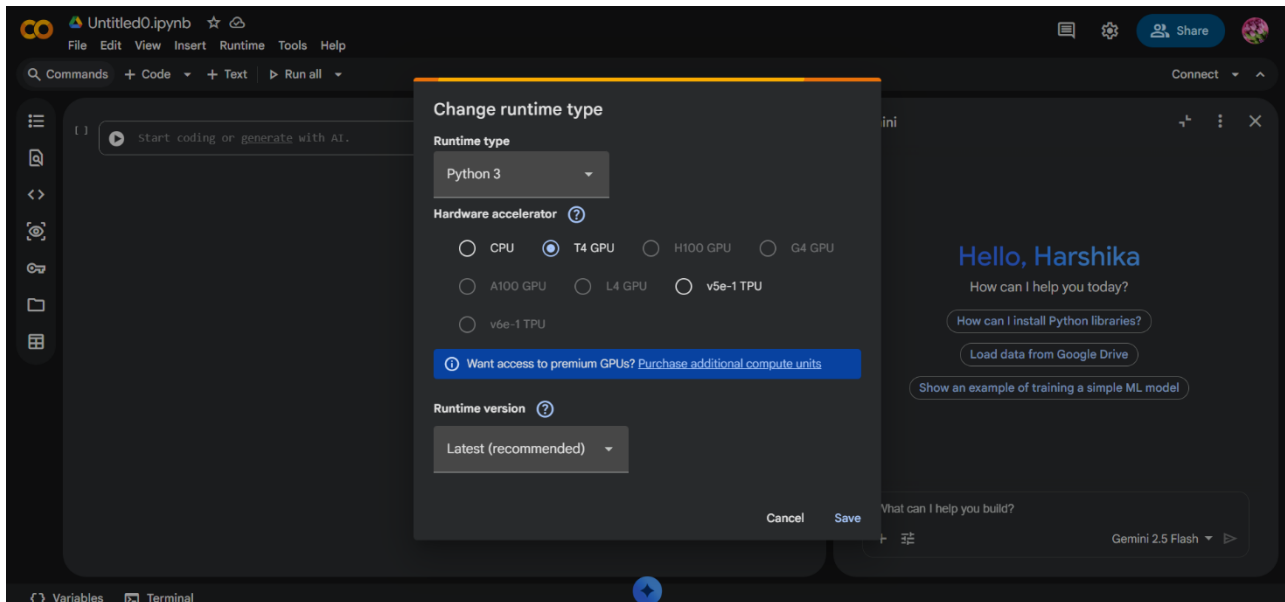
Content generated using this Space must comply with the official Hugging Face content policy (<https://huggingface.co/content-policy>).

This demo is for research and experimentation only. Generated production content should **not** be used for commercial purposes.

6. Create an account in <https://colab.research.google.com>

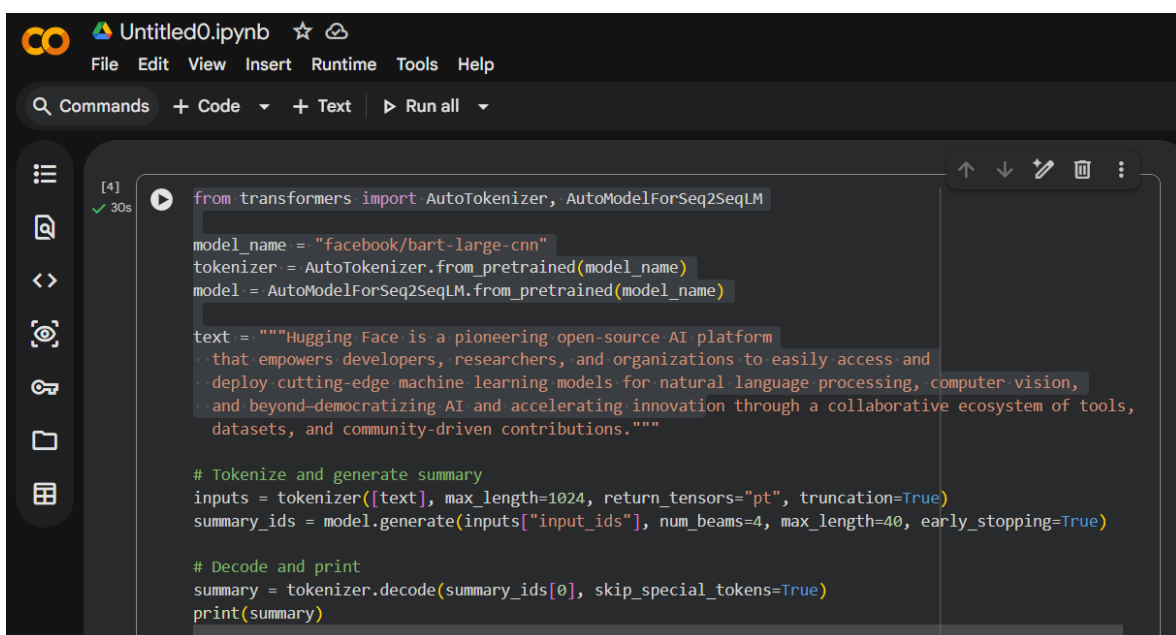


7. Change Runtime to GPU



9. Run the summarization example in a codeblock, here the Hugging Face open model will be downloaded to Google Colab and then it will be executed. Because of this it will take some time to run the model for the first time.

<https://colab.research.google.com/drive/1yc0uA5hzYCWcan5xdWrJB-c5EjJ6PIIv>

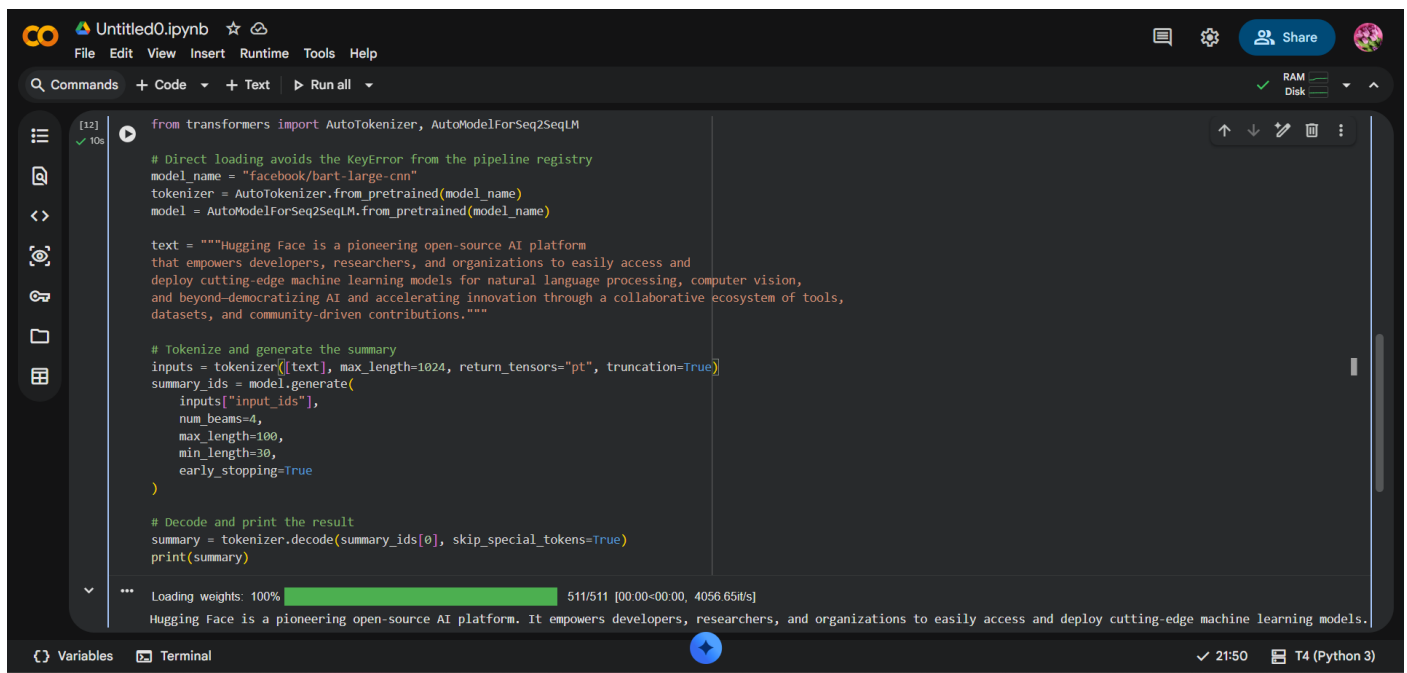


```
... vocab.json: 100% ██████████ 899k/899k [00:00<00:00, 22.0MB/s]
merges.txt: 100% ██████████ 456k/456k [00:00<00:00, 24.4MB/s]
tokenizer.json: 100% ██████████ 1.36M/1.36M [00:00<00:00, 45.6MB/s]
model.safetensors: reconstructing file: 100% ██████████ 1.63GB / 1.63GB, 95.6MB/s
model.safetensors: downloading bytes: ██████████ 1.09GB, 80.1MB/s
[transformers] Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights: 100% ██████████ 511/511 [00:00<00:00, 9661.77it/s]
generation_config.json: 100% ██████████ 363/363 [00:00<00:00, 38.1kB/s]
/usr/local/lib/python3.13/dist-packages/transformers/generation/utils.py:1700: UserWarning: Unfeasible l
warnings.warn(
Hugging Face is a pioneering open-source AI platform. It empowers developers, researchers, and organizat
```

10. Have a look at a specific summarization model available in HuggingFace
- <https://huggingface.co/EbanLee/kobart-summary-v3>

The screenshot shows the Hugging Face interface for the model **EbanLee/kobart-summary-v3**. The page includes a search bar, model tags (Summarization, Transformers, Safetensors, Korean, bart), and a text2text-generation label. It features a 'Model card' section with a line graph showing 'Downloads last month' at 2,657. Below this, there are tabs for 'Safetensors' (Model size: 0.1B params, Tensor type: F32) and 'Inference Providers' (HF Inference API). At the bottom, there is a 'Summarization' section with a text input field labeled 'Your prompt here...'.

10. Run the summarization for this model given below, here the Hugging Face open model will be downloaded to Google Colab and then it will be executed. Because of this it will take some time to run the model for the first time.



```
[12] from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Direct loading avoids the KeyError from the pipeline registry
model_name = "facebook/bart-large-cnn"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

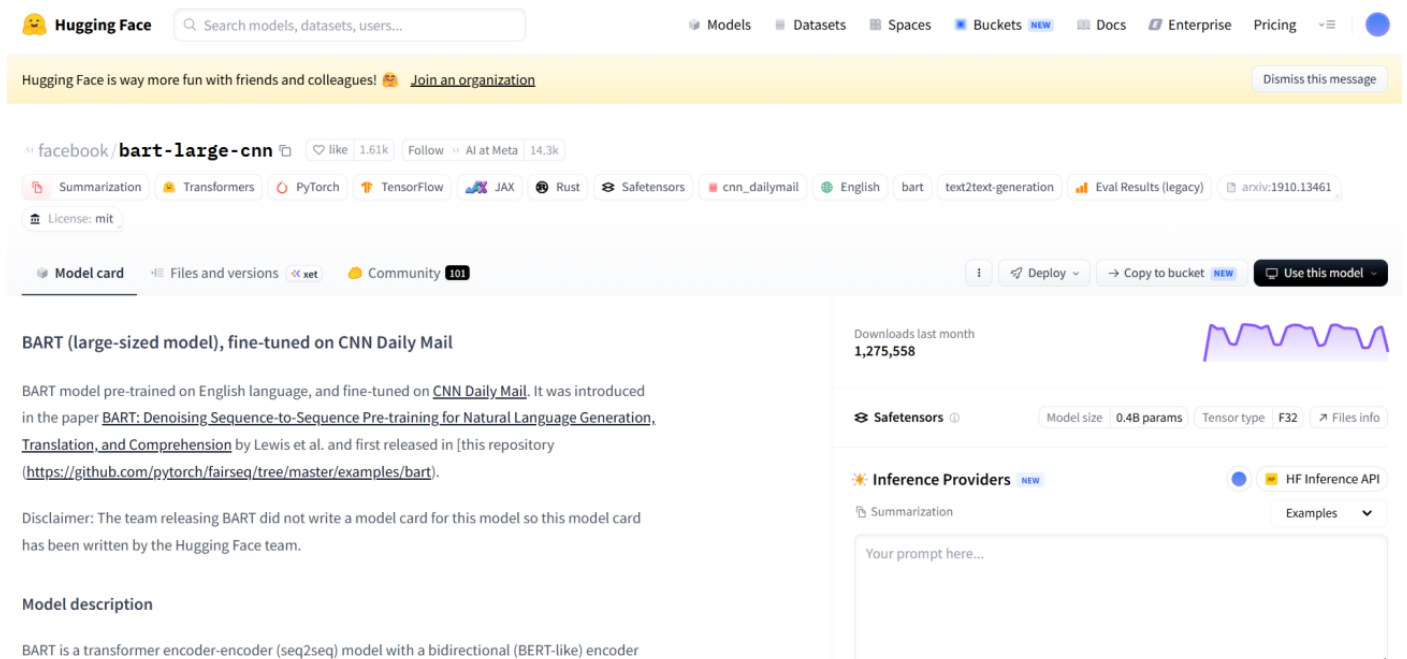
text = """Hugging Face is a pioneering open-source AI platform
that empowers developers, researchers, and organizations to easily access and
deploy cutting-edge machine learning models for natural language processing, computer vision,
and beyond—democratizing AI and accelerating innovation through a collaborative ecosystem of tools,
datasets, and community-driven contributions."""

# Tokenize and generate the summary
inputs = tokenizer([text], max_length=1024, return_tensors="pt", truncation=True)
summary_ids = model.generate(
    inputs["input_ids"],
    num_beams=4,
    max_length=100,
    min_length=30,
    early_stopping=True
)

# Decode and print the result
summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
print(summary)

*** Loading weights: 100% 511/511 [00:00<00:00, 4056.65it/s]
Hugging Face is a pioneering open-source AI platform. It empowers developers, researchers, and organizations to easily access and deploy cutting-edge machine learning models.
```

12. Select a Natural Language Model of your choice from Hugging Face



Hugging Face Search models, datasets, users...

Hugging Face is way more fun with friends and colleagues! [Join an organization](#)

facebook/**bart-large-cnn** like 1.61k Follow AI at Meta 14.3k

Summarization Transformers PyTorch TensorFlow JAX Rust Safetensors cnn_dailymail English bart text2text-generation Eval Results (legacy) arxiv:1910.13461

License: mit

Model card Files and versions xet Community 101

BART (large-sized model), fine-tuned on CNN Daily Mail

BART model pre-trained on English language, and fine-tuned on [CNN Daily Mail](#). It was introduced in the paper [BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension](#) by Lewis et al. and first released in [this repository (<https://github.com/pytorch/fairseq/tree/master/examples/bart>)].

Disclaimer: The team releasing BART did not write a model card for this model so this model card has been written by the Hugging Face team.

Model description

BART is a transformer encoder-encoder (seq2seq) model with a bidirectional (BERT-like) encoder

Downloads last month 1,275,558

Safetensors Model size 0.4B params Tensor type F32 Files info

Inference Providers HF Inference API

Summarization Examples

Your prompt here...

13. Try your best to see if you can get your selected Model running in Google CoLab. In some instances you will have to install additional python libraries. Some models have on the Right Side called Use this Model, where you can get Google CoLab code that you can try to use.



The screenshot shows the Google Colab interface for a notebook titled 'bart-large-cnn.ipynb'. The top bar includes the Colab logo, a 'Cannot save changes' warning, and a 'Share' button. Below the top bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar and tabs for 'Commands', 'Code', 'Text', 'Run all', and 'Copy to Drive' are also present. The left sidebar contains icons for file management and runtime settings. The main content area is titled 'Local Inference on GPU' and contains the following text:

Model page: <https://huggingface.co/facebook/bart-large-cnn>

⚠️ If the generated code snippets do not work, please open an issue on either the [model repo](#) and/or on [huggingface.js](#) 🙏

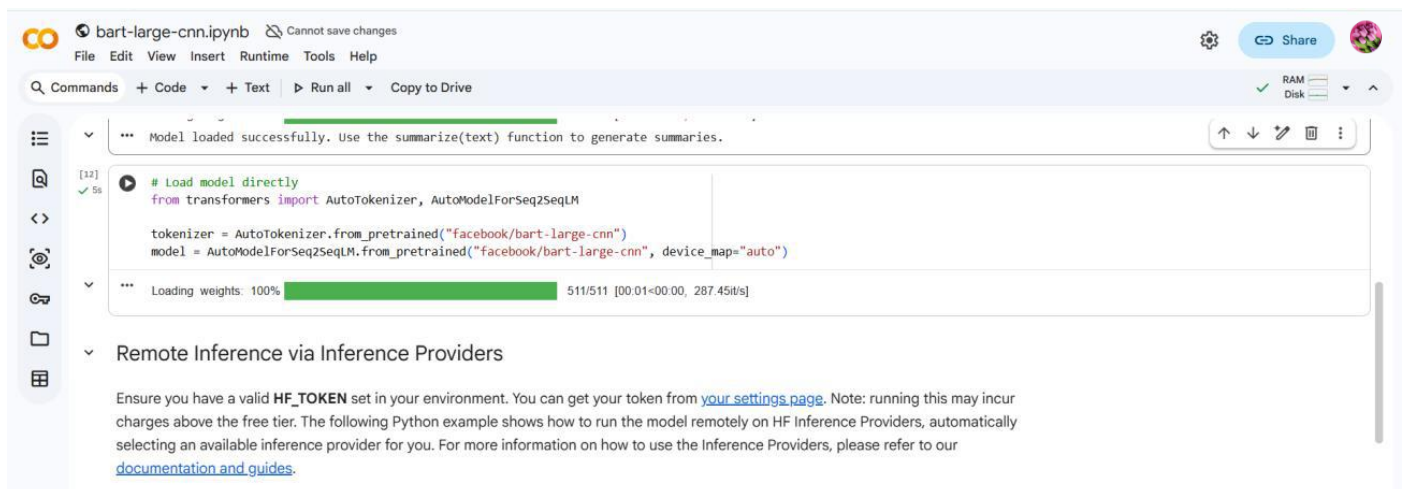
```
[11] ✓ 3s
# Load model directly and use manual generation instead of the pipeline task
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
import torch

model_name = "facebook/bart-large-cnn"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

# Define a helper function for summarization
def summarize(text):
    inputs = tokenizer([text], max_length=1024, return_tensors="pt", truncation=True)
    summary_ids = model.generate(inputs["input_ids"], num_beams=4, max_length=150, early_stopping=True)
    return tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print("Model loaded successfully. Use the summarize(text) function to generate summaries.")
```

Below the code, a progress bar shows 'Loading weights: 100%' with a green bar. The text '511/511 [00:00<00:00, 1278.96it/s]' is displayed. Below the progress bar, the text 'Model loaded successfully. Use the summarize(text) function to generate summaries.' is shown. The bottom of the screenshot shows the start of the next code cell [12].



The screenshot shows the Google Colab interface for a notebook titled 'bart-large-cnn.ipynb'. The top bar includes the Colab logo, a 'Cannot save changes' warning, and a 'Share' button. Below the top bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar and tabs for 'Commands', 'Code', 'Text', 'Run all', and 'Copy to Drive' are also present. The left sidebar contains icons for file management and runtime settings. The main content area is titled 'Remote Inference via Inference Providers' and contains the following text:

Ensure you have a valid **HF_TOKEN** set in your environment. You can get your token from [your settings page](#). Note: running this may incur charges above the free tier. The following Python example shows how to run the model remotely on HF Inference Providers, automatically selecting an available inference provider for you. For more information on how to use the Inference Providers, please refer to our [documentation and guides](#).

```
[12] ✓ 5s
# Load model directly
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn", device_map="auto")
```

Below the code, a progress bar shows 'Loading weights: 100%' with a green bar. The text '511/511 [00:01<00:00, 287.45it/s]' is displayed. The bottom of the screenshot shows the start of the next code cell [13].